fingerTips

Instruction:

- ***Use the csv files for your analysis***

- ***Time Duration 4 hours***

- ***Python, ML , SQL and Tableau Evaluation***

- ***Only after cleaning the data in Python, learners are allowed to move forward with Machine Learning, SQL, and Tableau tasks***

- ***Directly using the dataset in SQL, Tableau, or ML without cleaning it in Python will be considered noncompliance***

****Note –Use all csv dataset to Answer below questions***

Python Questions

1. Identify the maximum and minimum values of the selling_price column.
   Calculate the average of these two values and use this average to fill all missing values in selling_price.

2. Handle the km_driven and owner_count columns using an appropriate real-world data-cleaning technique.
   Justify the method used for each column based on data distribution and practical usage.

3. For the km_category column, 50% of the missing values should be filled with the most frequently occurring category.
   The remaining 50% of missing values should be filled with the value "High".

4. Fill missing values in the price_category column using the most frequently occurring category in the dataset.
   If it is not possible to determine whether the category is generated manually or automatically, replace missing values with "Unknown".

5. For the fuel column, distribute all missing values equally among the existing unique fuel types.
   Each unique fuel value must be assigned the same number of missing records.

6. After completing all the data cleaning steps, split the final cleaned dataset into two CSV files and export them.

   First file =>
   Name it: vehicle_info

   This file should contain only the following columns:
   vehicle_uid, name, year, fuel, seller_type, transmission, owner, brand, car_age, etc.

   Second file=>
   Name it: vehicle_sales

This file should contain only the following columns: vehicle_uid, selling_price, km_driven, price_category, km_category, etc.

Important Note: Both datasets should have the same file name as mentioned in Question-6, else your answers will not be evaluated.

SQL Questions

Q1. Show all vehicles that use Diesel fuel.

Q2. Display vehicle name, year, and brand for all vehicles.

Q3. Find all sales where selling price is below 5,00,000.

Q4. Find fuel types where average selling price is greater than 6,00,000.

Q5. Find brands that have more than 50 vehicles listed.

Q6. Show transmission types where the average selling price is greater than 7,00,000 AND the total number of vehicles is at least 30.

Q7. Find price categories where the average km driven is above 75,000, but only include categories that have more than 20 vehicles AND only count Diesel vehicles.

Q8. Find vehicles whose selling price is higher than the overall average selling price.

Q9. Find brands whose average selling price is greater than the average selling price of Toyota vehicles.

Q10. Find fuel types where total selling price is higher than total selling price of Petrol vehicles.

Tableau Evaluation

- *Create a dashboard and use all the insights that you can create using the dataset and use filter and action.*
- *Format all the sheets with filters and dashboard.*
- *Create a story for the sheets and dashboard that you have created.*

ML Evaluation

Q1. Load the cleaned Car Evaluation dataset. Identify the Type of ML Problem. Select 'price_category' as target variable, and print the shape of X and y.

Q2. Identify categorical columns, apply Encoding techniques (if necessary), perform Standard Scaling on numerical features.

Q.3.Plot the distribution of car price categories. Analyse the relationship between safety, selling price indicators, and number of owners or usage (km driven) with the target variable price category

Q.4 Split the dataset into training and testing sets. Train a SVM model and display accuracy Score, Confusion Matrix, Classification report.

Q.5. Train a Random Forest Classifier on the same data and predict price_category, find - accuracy Score, Confusion Matrix, Classification report. Compare both models accuracy and print the best model for Prediction.

Good Luck & Happy Learning